

6406534471085. ✖ ('e', 'r')

6406534471086. ✔ ('r', '</w>')

6406534471087. ✖ ('b', 'r')

6406534471088. ✖ ('r', 'i')

AI

Section Id :	64065394965
Section Number :	15
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653217864
Question Shuffling Allowed :	No

Question Number : 347 Question Id : 6406531338249 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534471089. ✓ YES

6406534471090. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653217865

Question Shuffling Allowed :

No

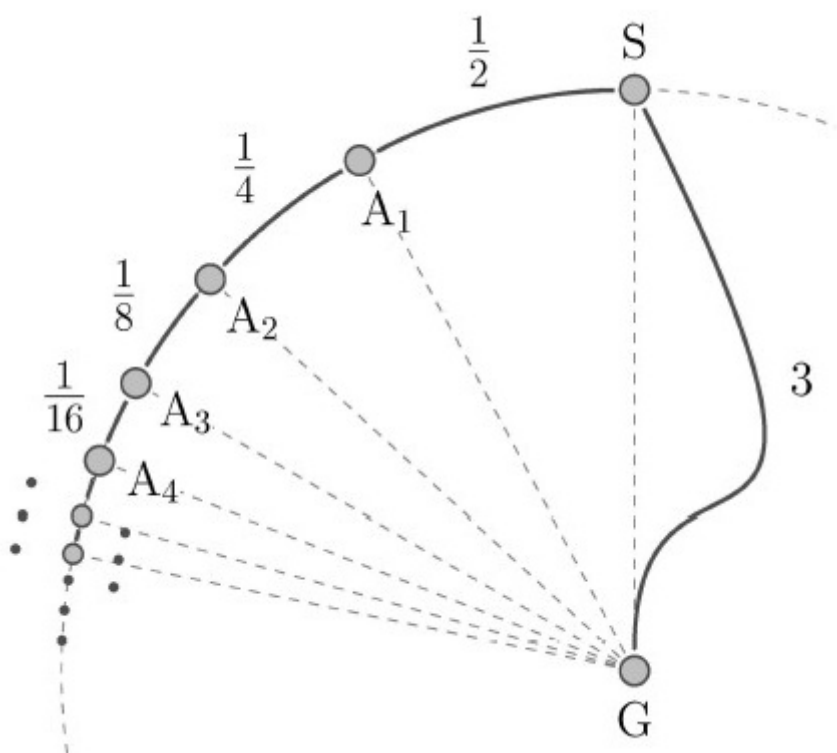
Question Id : 6406531338250 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (348 to 353)

Question Label : Comprehension

SEARCH

Figure shows an infinite state space with finite branching, where S is the start node and G is the goal node, and G is at the center of a unit circle, and the remaining nodes are on the unit circle. (Note: circular grid is shown by dashed lines which are not part of the state space.)



Take the radial distance as the heuristic value:

$h(G)=0$; $h(S)=1$; and $h(A_k)=1$, for all k in $\{1, 2, \dots\}$.

MoveGen returns neighbours in alphabetical order (A1,A2,..., G,S).

Tie-breaker: use alphabetical order to break ties.

Note: $\text{sum}(1, a, a^2, a^3, \dots, a^{(n-1)})$ is given by $((1 - a^n)/(1 - a))$.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 348 Question Id : 6406531338251 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the Breadth First Search find the path to the goal?

Options :

6406534471091. ✓ Yes

6406534471092. ✗ No

6406534471093. ✗ Cannot be determined

Question Number : 349 Question Id : 6406531338252 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the Best First Search find the path to the goal?

Options :

6406534471094. ✓ Yes

6406534471095. ✗ No

6406534471096. ✗ Cannot be determined

Question Number : 350 Question Id : 6406531338253 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does the Branch and Bound algorithm find the path to the goal?

Options :

6406534471097. ✗ Yes

6406534471098. ✓ No

6406534471099. ✗ Cannot be determined

Question Number : 351 Question Id : 6406531338254 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does A* algorithm find the path to the goal?

Options :

6406534471100. ✗ Yes

6406534471101. ✓ No

6406534471102. ✗ Cannot be determined

Question Number : 352 Question Id : 6406531338255 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Does wA^* ($w=3$) algorithm find the path to the goal?

Options :

6406534471103. ✓ Yes

6406534471104. ✗ No

6406534471105. ✗ Cannot be determined

Question Number : 353 Question Id : 6406531338256 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the heuristic admissible in the given state space?

Options :

6406534471106. ✓ Yes

6406534471107. ✗ No

6406534471108. ✗ Cannot be determined

Sub-Section Number :

3

Sub-Section Id :

640653217866

Question Shuffling Allowed :

No

Question Id : 6406531338257 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (354 to 358)

Question Label : Comprehension

TSP

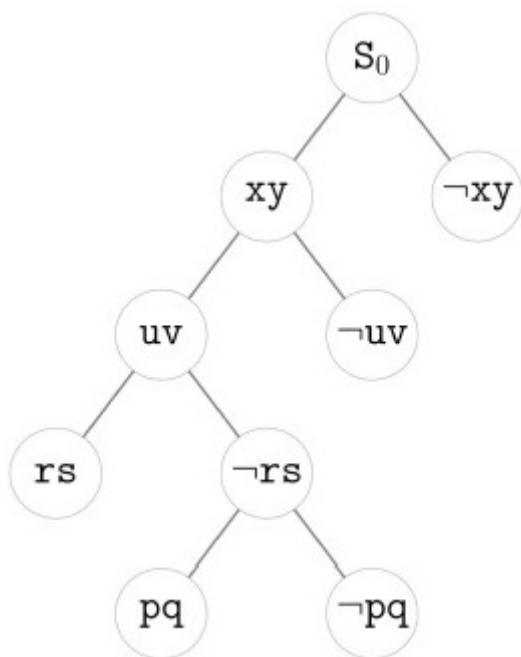
The distance matrix for 5 cities (A to E) and the sorted edge list is provided below.
For each city the distances to other cities are listed in ascending order.
For example, in row one, the distance from A to B is 30, A to C is 32 and so on.

A	B:30	C:32	D:68	E:70
B	A:30	C:36	E:40	D:44
C	A:32	B:36	D:50	E:72
D	E:42	B:44	C:50	A:68
E	B:40	D:42	A:70	C:72

AB	AC	BC	BE	DE
30	32	36	40	42

BD	CD	AD	AE	CE
44	50	68	70	72

The search tree traversed by TSP BnB shows four segments. Identify the four segments xy , uv , rs , pq and then answer the given sub-questions.



Attention: A segment is a two-way edge between two cities. After adding/dropping a permanent segment, infer as many new permanent segments as possible before computing the lower bound.

Sub questions

Question Number : 354 Question Id : 6406531338258 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the lower bound on the cost of the tour as per the TSP BnB algorithm discussed in class?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 17

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

182

Question Number : 355 Question Id : 6406531338259 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the first segment (XY) selected for refining S0?

Enter edge label XY

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: XY

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

AB

BA

Question Number : 356 Question Id : 6406531338260 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the cost of the node ~XY (that permanently excludes segment XY)?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

206

Question Number : 357 Question Id : 6406531338261 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

How many tours are represented in the node rs?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 358 Question Id : 6406531338262 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the cost of the node ~pq (that permanently excludes segment pq)?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 17

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

320

Sub-Section Number :

4

Sub-Section Id :

640653217867

Question Shuffling Allowed :

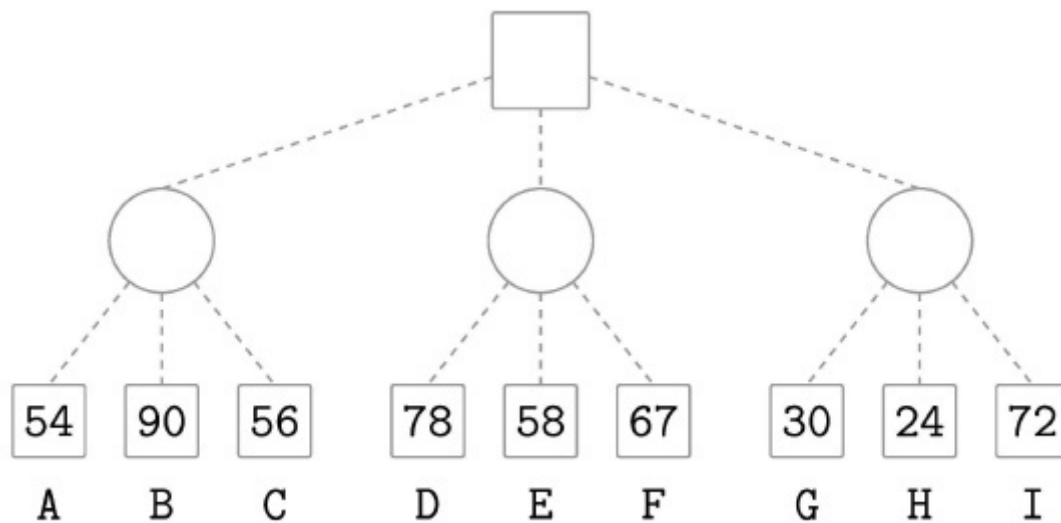
No

Question Id : 6406531338263 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (359 to 365)

Question Label : Comprehension

GAMES



Based on the above data, answer the given subquestions

Sub questions

Question Number : 359 Question Id : 6406531338264 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes in the best strategy.

Enter the node labels as a comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

D,E,F

Question Number : 360 Question Id : 6406531338265 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes pruned by Alpha-Beta algorithm.

Enter the node labels as a comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

H,I

Question Number : 361 Question Id : 6406531338266 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes in the initial cluster formed by SSS* algorithm.

Enter the node labels as a comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,D,G

Question Number : 362 Question Id : 6406531338267 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Solve the game tree using SSS* algorithm.

List the horizon nodes (not in the initial cluster) that are assigned SOLVED status.

Enter the node labels as a comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

E,F

Question Number : 363 Question Id : 6406531338268 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements about the Alpha-Beta algorithm.

Options :

6406534471118. ✓ Alpha-Cut prunes MAX nodes.

6406534471119. ✓ Alpha-Cut is made by a MIN node.

6406534471120. ✓ Beta-Cut prunes MIN nodes.

6406534471121. ✓ Beta-Cut is made by a MAX node.

Question Number : 364 Question Id : 6406531338269 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Select the condition that triggers an Alpha-Cut.

Options :

6406534471122. ✓ $\text{eval}(\text{child}) \leq \text{Alpha}$

6406534471123. ✗ $\text{Alpha} < \text{eval}(\text{child}) < \text{Beta}$

6406534471124. ✗ $\text{Beta} \leq \text{eval}(\text{child})$

6406534471125. ✗ Cannot be determined

Question Number : 365 Question Id : 6406531338270 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For a 2-ply game tree with branching factor 3, the number of strategies for MAX-player is _____.

Options :

6406534471126. ✗ 2

6406534471127. ✓ 3

6406534471128. ✗ 2^2

6406534471129. ✗ 2^3

6406534471130. ✗ 3^2

6406534471131. ✗ 3^3

Sub-Section Number :

5

Sub-Section Id :

640653217868

Question Shuffling Allowed :

No

Question Id : 6406531338271 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (366 to 372)

Question Label : Comprehension

AUTOMATED PLANNING

The domain description of a Blocks World with a single one-armed robot is given below.

Note: this is the same domain description used in the weekly assignments.

PREDICATES

armEmpty	The arm is not holding any block, it is empty.
holding(X)	The arm is holding X.
onTable(X)	X is on the table.
clear(X)	X has nothing above it, it is clear.
on(X,Y)	X is directly placed on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add Effects : { holding(X) }

Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.

Preconditions: { holding(X) }

Add Effects : { armEmpty, onTable(X) }

Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }

Add Effects : { clear(Y), holding(X) }

Del Effects : { armempty, on(X,Y) }

Stack(X,Y): place X directly on top of Y.

Preconditions: { holding(X), clear(Y) }

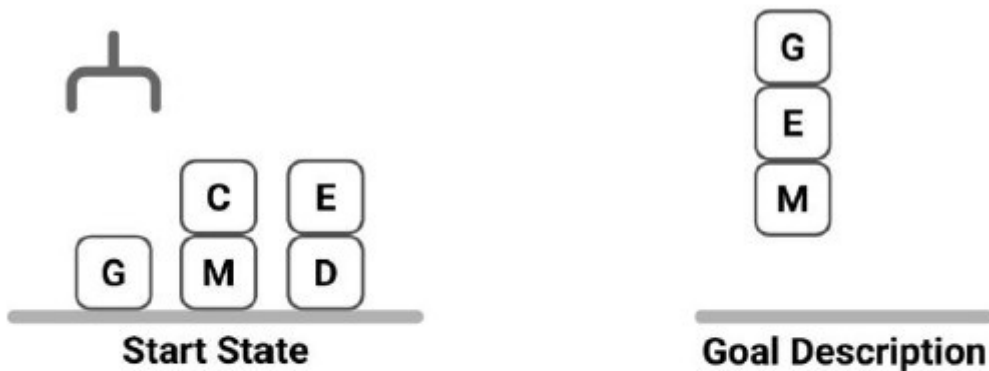
Add Effects : { armEmpty, on(X,Y) }

Del Effects : { holding(X), clear(Y) }

Tie-breaker: When actions are chosen non-deterministically, choose actions that lead to a plan. Throw away the actions that lead to dead-ends or cycles.

Tie-breaker: Treat the goal description, preconditions and effects as lists that are accessed from left to right. When the elements in a list are pushed one by one to a stack, the last element in the list will be at the top of the stack. It has the effect of reversing the list.

A planning problem is given below, find a plan using the operators and predicates defined in the blocks-world domain.



Start: { armEmpty, onTable(G), onTable(M), onTable(D),
clear(G), clear(C), clear(E), on(C,M), on(E,D) }

Goal Description: { on(E,M), on(G,E) }

Based on the above data, answer the given subquestions

Sub questions

Question Number : 366 Question Id : 6406531338272 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the length of the optimal plan? Enter the number of actions in the optimal plan.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 367 Question Id : 6406531338273 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions for the given planning problem?

Options :

6406534471133. ✖ Pickup(D)

6406534471134. ✔ Pickup(G)

6406534471135. ✖ Pickup(M)

6406534471136. ✔ Unstack(C,M)

6406534471137. ✔ Unstack(E,D)

Question Number : 368 Question Id : 6406531338274 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **relevant** actions for the given planning problem?

Options :

6406534471138. ✖ Pickup(G)

6406534471139. ✔ Stack(E,M)

6406534471140. ✔ Stack(G,E)

6406534471141. ✖ Unstack(C,M)

6406534471142. ✖ Unstack(E,D)

Question Number : 369 Question Id : 6406531338275 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following can be pushed as the first three elements onto the stack by the Goal Stack Planning algorithm? In the representation below the stack has a bottom on the right marked by the entry END. Use appropriate tie-breakers listed in the main data.

Options :

6406534471143. ✖ { on(E,M), on(G,E) }; on(E,M); on(G,E); END

6406534471144. ✖ { on(E,M), on(G,E) }; on(G,E); on(E,M); END

6406534471145. ✔ on(G,E); on(E,M); { on(E,M), on(G,E) }; END

6406534471146. ✖ on(E,M); on(G,E); { on(E,M), on(G,E) }; END

Question Number : 370 Question Id : 6406531338276 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the subgoal ordering given in the goal description (and using the given tie breaking rules), which of the following is the first action popped out of the stack in Goal Stack Planning?

Options :

6406534471147. ✔ Pickup(G)

6406534471148. ✖ Stack(G,E)

6406534471149. ✖ Unstack(E,D)

6406534471150. ✖ Unstack(C,M)

Question Number : 371 Question Id : 6406531338277 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the subgoal ordering given in the goal description (and using the given tie breaking rules), does the Goal Stack Planning algorithm find an optimal plan?

Options :

6406534471151. ✖ Yes

6406534471152. ✔ No

6406534471153. ✖ Cannot be determined

Question Number : 372 Question Id : 6406531338278 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the subgoals in the goal description are swapped places then does the Goal Stack Planning algorithm find an optimal plan?

Options :

6406534471154. ✔ Yes

6406534471155. ✖ No

6406534471156. ✖ Cannot be determined